

ASWIN JOJO

aswintharakkunnel@gmail.com | (832) 427-8618 | [LinkedIn](#) | [GitHub](#)

SUMMARY

Full-Stack Software Engineer with ~3 years of experience building and scaling distributed web applications across **React, Node.js, Python, FastAPI, PostgreSQL, AWS**. Proven track record delivering high-impact systems, including a lead management platform serving **70M+ records**, low-latency APIs optimized from **~3s to ~200ms**, and cloud-native applications integrating data and machine learning workflows.

PROFESSIONAL EXPERIENCE

Software Engineer, University of Houston | Houston, TX

Sep '25 - Present

Skills - Python, FastAPI, PostgreSQL, Next.js, React

- Designed **FastAPI services** and PostgreSQL query pipelines powering real-time analytics across **30,000+ U.S. zip codes**, optimizing server-side aggregation to eliminate client-side computation and reduce data transfer overhead.
- Built a geospatial analytics platform integrating mobility datasets, ML inference outputs, and satellite-derived features into a unified **Next.js application** used for large-scale market intelligence analysis.

Software Engineer Intern, Resolve Tech Solutions | Dallas, TX

Jun '25 - Aug '25

Skills - React, FastAPI, MySQL, AWS, Microservices, Redux, CI/CD, Machine Learning, REST APIs, GitLab, EC2

- Architected and delivered a **React/FastAPI onboarding platform** supporting **multi-tenant user provisioning across 20+ clients**, designing **role-based workflows** and service integrations that **eliminated manual onboarding dependencies for 400+ employees**.
- Built an automated ticket assigning pipeline integrating ML inference, ServiceNow workflows, and human feedback loops, **reducing average response time from 3 days to 1 day** while improving ticket routing accuracy.
- Implemented **Docker-based** containerized deployments and built **GitLab CI/CD pipelines automating build, test, and release stages**, reducing manual deployment effort and improving deployment consistency across **AWS EC2 environments**.

Software Engineer, Policybazaar.com | Gurgaon, India

Oct '22 - Jul '24

Skills - Node.js, React, Next.js, Typescript, MySQL, MongoDB, AWS SQS, Redis, REST APIs, SDLC, Agile, AI ChatBot

- Designed and delivered a **Lead Management Dashboard serving 70M+ leads**, building a **GraphQL + React/TypeScript** frontend and **Node.js** backend that reduced data load times and improved responsiveness for high-volume operations teams.
- Built a **WhatsApp chatbot** automating policy queries, renewals, and claims processing through backend validation and messaging API integration, improving lead response time and contributing to a **15% increase in quarterly lead conversion revenue** compared to the previous quarter.
- Optimized backend performance through **Redis caching, PostgreSQL indexing**, query rewrites, and connection pooling, reducing API response time and query execution from ~3s to ~200ms across Lead and Claims services.
- Architected **event-driven REST APIs** using **AWS SQS** for asynchronous message processing, decoupling lead ingestion from downstream services to handle traffic spikes without data loss or cascading failures.
- Owned **end-to-end** development of **Claims Management** and **Renewal Portal** features, scalable Node.js services and React interfaces reducing claims turnaround time by 50% and eliminating manual workflow bottlenecks.

PROJECTS

GenAI News Aggregator | Python, PostgreSQL, OpenAI API, Docker, SQLAlchemy, GitHub Actions

Feb '26

- Built a nightly **multi-agent pipeline** orchestrating prompt chains for article scraping, LLM-based summarization, and an **LLM-as-judge** ranking layer that scores content quality and relevance before delivery.
- Implemented **per-user interest profiling** stored in PostgreSQL to personalize digest ranking and content selection; automated end-to-end pipeline via cron with Docker-containerized deployment and GitHub Actions CI/CD.

Population Mobility Intelligence Platform | Python · FastAPI · PostgreSQL · Scikit-learn · GeoPandas · React

Sep '25

- Engineered end-to-end ETL pipelines processing **50,000+ GeoTIFF files** from Sentinel-2 satellite imagery and Veraset mobility data, consolidating visit frequency, dwell time, and peak-hour signals per zip code into a unified PostgreSQL schema.
- Trained a logistic regression classifier on engineered features (greenness indices, visit density, seasonal signals) to categorize zip codes by population movement potential, achieving **85% accuracy vs. a 61% baseline**, directly informing marketing zone prioritization

CERTIFICATIONS & ACHIEVEMENTS

- Microsoft Certified:** Azure Fundamentals – AZ900
- AWS Solutions Architect Associate** – SAA
- Best Paper Award, **IEEE ICICICT 2022** – Intruder Detection System using IoT with Adaptive Face Monitoring and Motion Sensing

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, Typescript, SQL, HTML

Database: PostgreSQL, MongoDB, MySQL, MSSQL, DynamoDB, SQLite

Libraries/Framework/Tools: React.js/Next.js/Redux, Node.js/Express, FastAPI, Git, Jenkins, JIRA, Postman, GitHub Actions, CI/CD

Cloud Technologies: AWS (Lambda, SQS, EC2, Amplify, Cognito), Microsoft Azure

Development Practices: Full-stack development, CI/CD, Automation, TDD, Design Patterns, Agile, SDLC, SOLID, and Code Reviews

EDUCATION

Masters in Computer Science - University of Houston | 3.8/4.0

Aug '24 - May '26

Bachelor of Technology in Computer Science and Engineering - VIT Chennai | 9.1/10

Jul '19 - May '23